Unit II: Plant Morphology and Taxonomy of Angiosperm

Chapter 5

Taxonomy and Systematic Botany

Learning Objectives

The learner will be able to,

- *Differentiate systematic botany from taxonomy.*
- *Explain the ICN principles and to discuss the codes of nomenclature.*
- *Compare the national and international herbaria.*
- *Appreciate the role of morphology, anatomy, cytology, DNA sequencing in relation to Taxonomy,*
- *Describe diagnostic features of families Fabaceae, Apocynaceae, Solanaceae, Euphorbiaceae, Musaceae and Liliaceae.*

Chapter Outline

5.1 Taxonomy and Systematics
5.2 Taxonomic Hierarchy
5.3 Concept of species – Morphological, Biological and Phylogenetic
5.4 International Code of Botanical Nomenclature
5.5 Taxonomic Aids
5.6 Botanical Gardens
5.7 Herbarium – Preparation and uses
5.8 Classification of Plants
5.9 Need for classification
5.10 Types of classification
5.11 Modern trends in taxonomy
5.12 Cladistics
5.13 Selected Families of Angiosperms

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Plants are the prime companions of human beings in this universe. Plants are the source of food, energy, shelter, clothing, drugs, beverages, oxygen and the aesthetic environment. Taxonomic activity of human is not restricted to living organisms alone. Human beings learn to identify, describe, name and classify food, clothes, books, games, vehicles and other objects that they come across in their life. Every human being thus is a taxonomist from the cradle to the grave.

Taxonomy has witnessed various phases in its early history to the present day modernization. The need for knowledge on plants had been realized since human existence, a man started utilizing plants for food, shelter and as curative agent for ailments.

Theophrastus (372 – 287 BC), the Greek Philosopher known as "**Father of Botany**". He named and described some 500 plants in his *"De Historia Plantarum"*. Later Dioscorides (62 – 127 AD), Greek physician, described and illustrated in his famous **"Materia medica"** and described about 600 medicinal plants. From 16th century onwards Europe has witnessed a major developments in the field of Taxonomy. Some of the key contributors include Andrea Caesalpino, John Ray, Tournefort, Jean Bauhin and Gaspard Bauhin. Linnaeus ***'Species Plantarum'*** (1753) laid strong foundation for the binomial nomenclature.

Taxonomy is no more classical morphology based discipline but become a dynamic and transdisciplinary subject, making use of many branches of botany such as Cell Biology, Physiology, Biochemistry, Ecology,

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